

MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati

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Tutorial Sheet No. 3

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Lectures 7-10: Higher-order differential equations, Introduction to power series

Attention students: Please keep in mind that it is not possible to discuss all problems during the tutorial. You can consider the problems not discussed as your practice problems.

1. Consider the differential equation $y'' - 5y' + 6y = 0$.
 - (i) Show that e^{2x} and e^{3x} are linearly independent solutions of this equation in $-\infty < x < \infty$.
 - (ii) Write the general solution of the given equation.
 - (iii) Find the solution that satisfies the conditions $y(0) = 2$, $y'(0) = 3$. Is this solution unique? Justify your answer.
2. If y_1 and y_2 are linearly independent solutions of $xy'' + 2y' + xe^xy = 0$, $0 < x < \infty$ and if $W(y_1, y_2)(1) = 2$, find the value of $W(y_1, y_2)(5)$.
3. Let $P(x), Q(x) \in C(I)$. Assume that the functions $y_1, y_2 \in C^2(I)$ are solutions of the differential equations $y'' + P(x)y' + Q(x)y = 0$ on an open interval I . Prove that
 - (a) if y_1 and y_2 are zero at the same point in I , then they cannot be a fundamental set of solutions on that interval;
 - (b) if y_1 and y_2 have a common point of inflection x_0 in I , then they cannot be a fundamental set of solutions on that interval.
4. Find the general solution of the following differential equations:
 - (a) $\frac{d^4y}{dx^4} + y = 0$
 - (b) $\frac{d^5y}{dx^5} - 2\frac{d^4y}{dx^4} + \frac{d^3y}{dx^3} = 0$
 - (c) $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$
 - (d) $\frac{d^5y}{dx^5} + 5\frac{d^4y}{dx^4} + 10\frac{d^3y}{dx^3} + 10\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + y = 0$
5. Solve the following initial-value problems:
 - (a) $y'' - 2y' + y = 2xe^{2x} + 6e^x$; $y(0) = 1$, $y'(0) = 0$
 - (b) $y''(x) + y(x) = 3x^2 - 4\sin x$, $y(0) = 0$, $y'(0) = 1$

6. Use the method of undermined coefficients to find a particular solution to the following differential equations:

(a) $y'' - 3y' + 2y = 2x^2 + 3e^{2x}$

(b) $y'' - 3y' + 2y = xe^{2x} + \sin x$

7. Use the method of variation of parameters to find a particular solution, and hence the general solution for the following differential equations:

(a) $y'' - 5y' + 6y = \cos(2x) + 1$

(b) $y'' - 5y' + 6y = e^{3x} - x^2$

8. Use the operator method to find a particular solution of the following ODEs:

(a) $y''' + y'' + y' + y = x^5 - 2x^2 + x$

(b) $y''' - 5y'' + 8y' - 4y = 3e^{2x}$

(c) $y'' - 3y' + 2y = 3 \sin 2x$

9. If p is not zero or a positive integer, show that the series

$$\sum_{n=1}^{\infty} \frac{p(p-1)(p-2) \cdots (p-n+1)}{n!} x^n$$

converges for $|x| < 1$ and diverges for $|x| > 1$.

10. We know that

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots$$

is valid for $|x| < 1$. Applying it, show that

(a) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

(b) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$

for $|x| < 1$.

11. Consider the differential equation $y' + y = 1$. Find its power series solution and recognize it as the expansion of a familiar function. Further, verify your power series solution by solving the ODE by an elementary method.

12. Repeat the above for the ODE $y' = x - y$. Use the initial condition $y(0) = 0$.

The End